Ex.No:09

Capture packet traces of ARP protocol using Wireshark

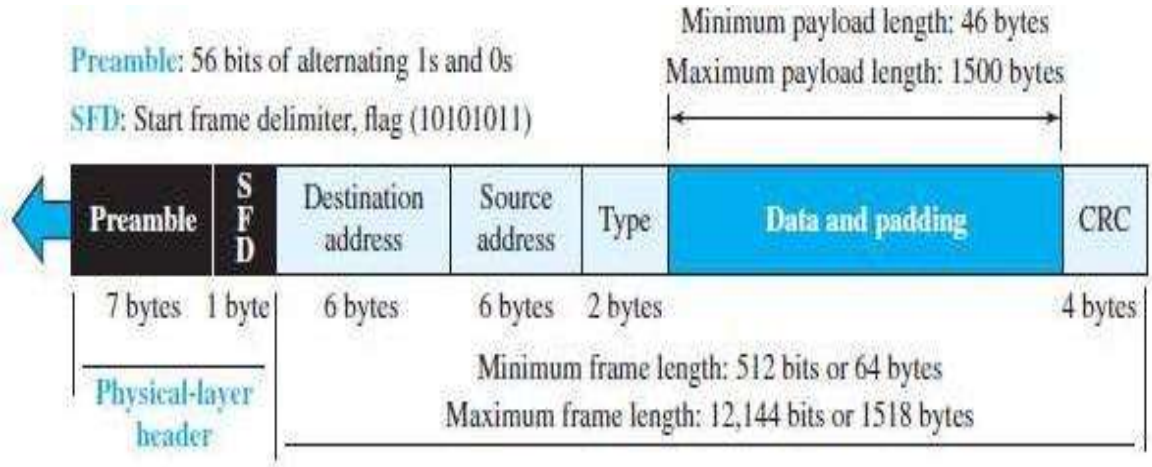
Aim

To capture and analyzes the packet traces of Ethernet and ARP protocol using Wireshark

Ethernet Protocol

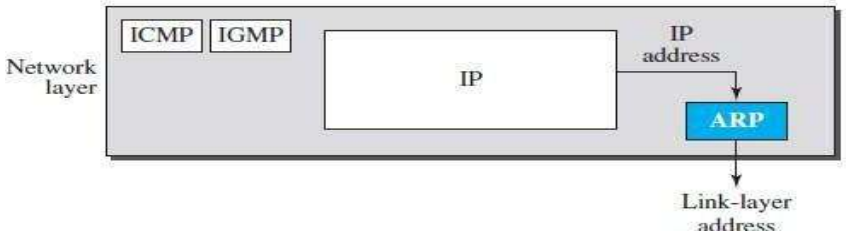
- Ethernet is a standard communication protocol used to connect computers within LAN and WAN
- Widely used wired LAN technology

Ethernet Frame Format:



ARP Protocol

- ARP accepts an IP address from the IP protocol, maps it to the corresponding link- layer address (MAC address) and passes it to the data-link layer.
- IP -> MAC address

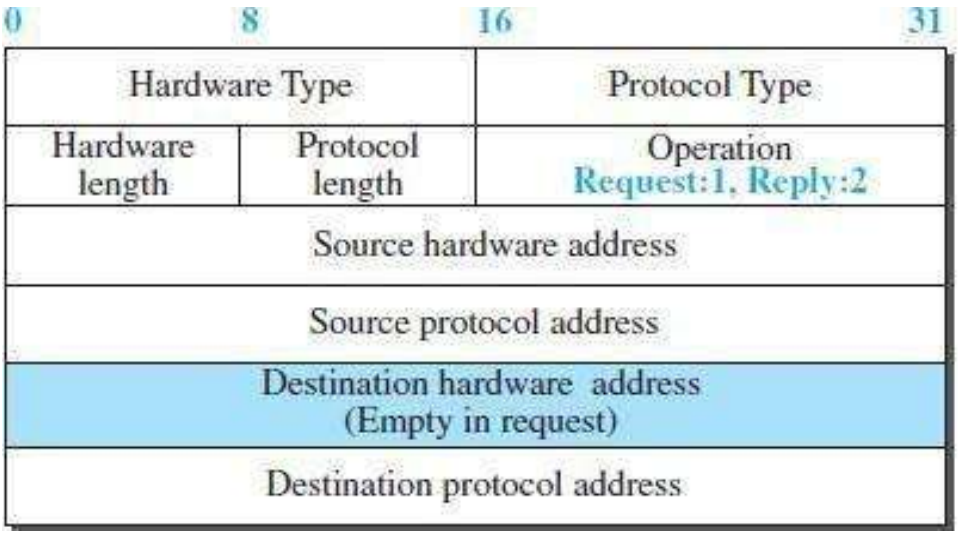


ARP packet format

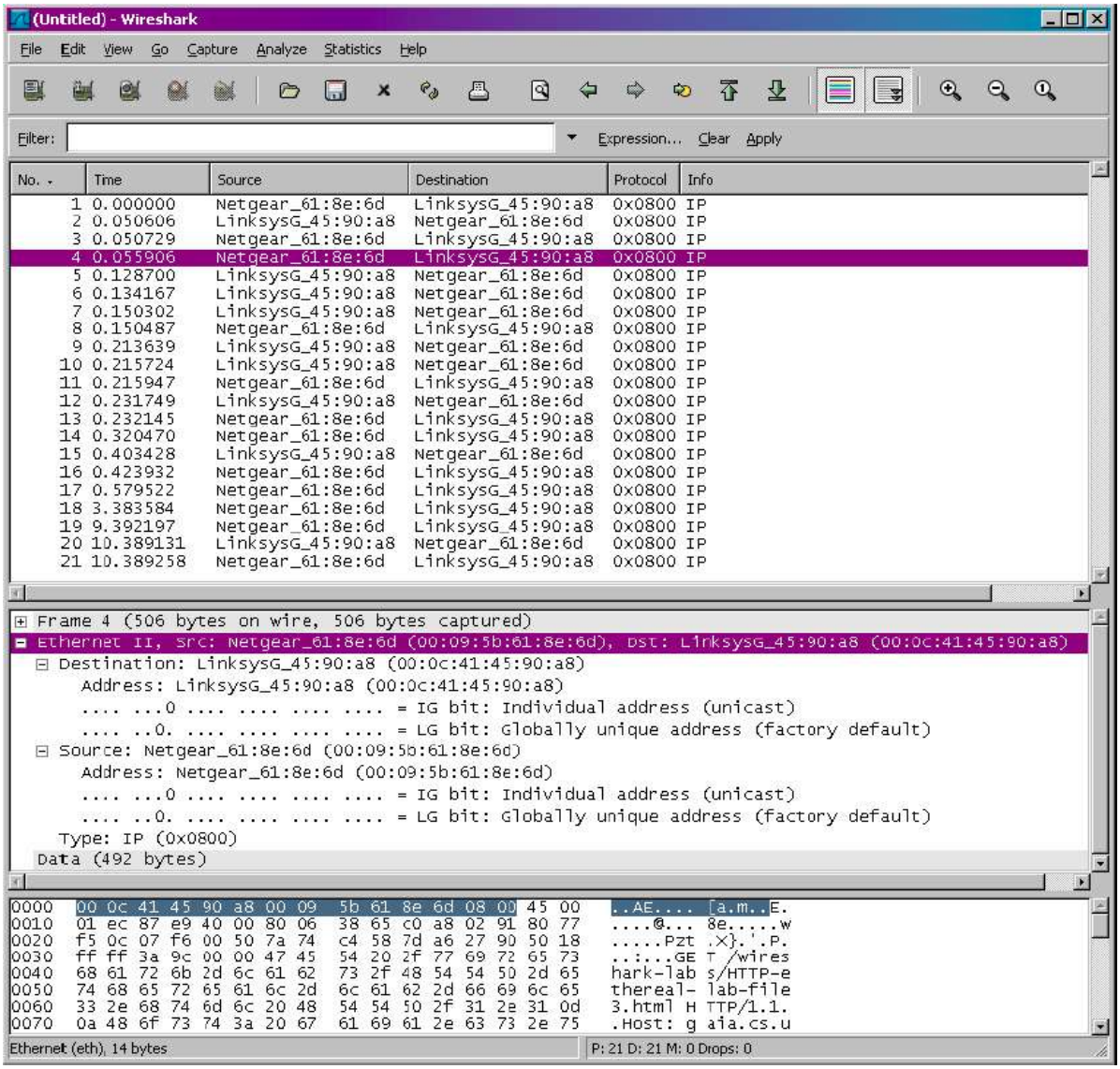
ARP request and response messages are send as follows:

- ARP Request: Broadcast
- ARP Reply: Unicast

Below diagram shows the packet format (both request and reply) of ARP protocol

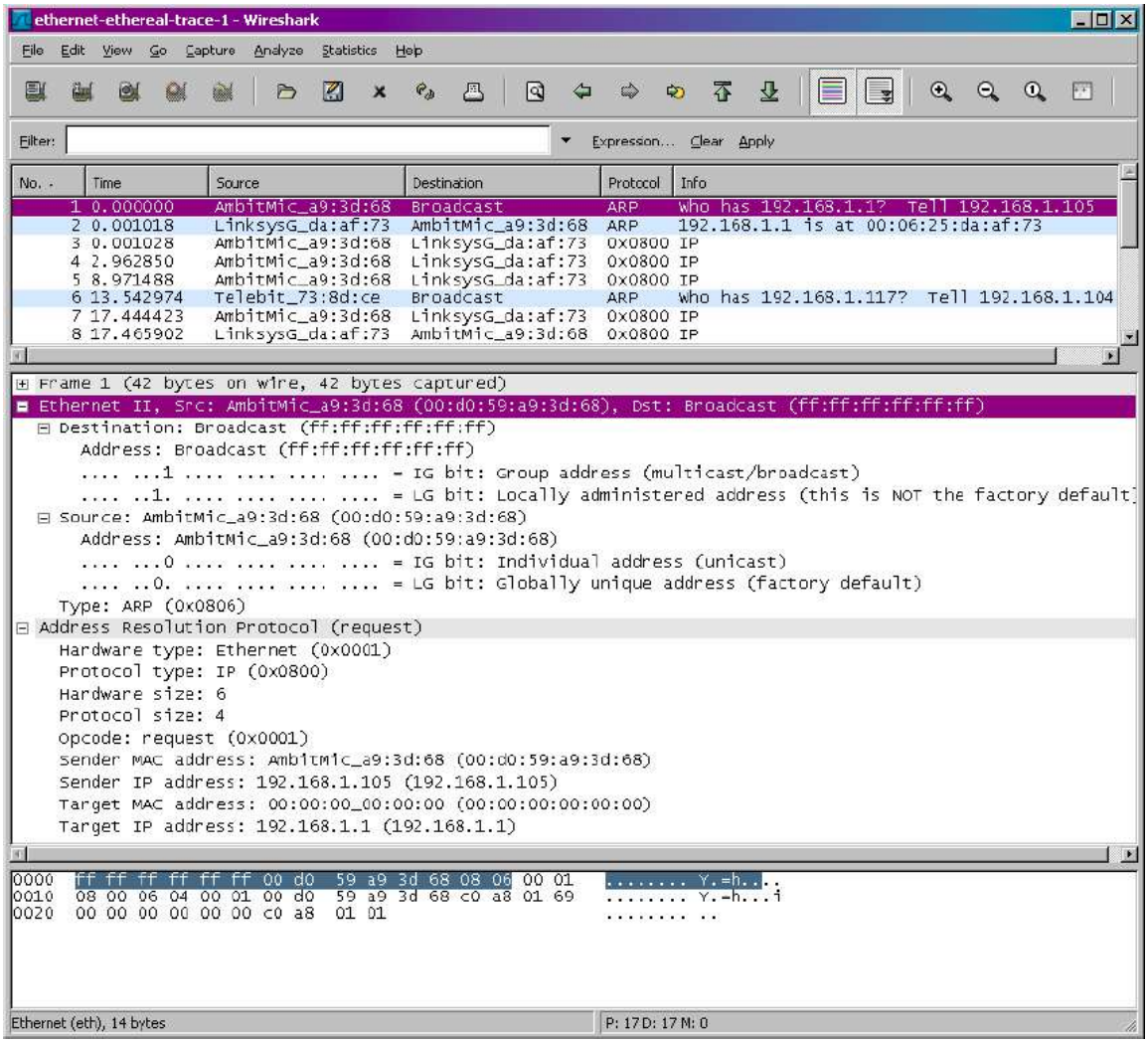


Shown below is the screen shot expanding the view of the Ethernet frame carrying the HTTP GET:



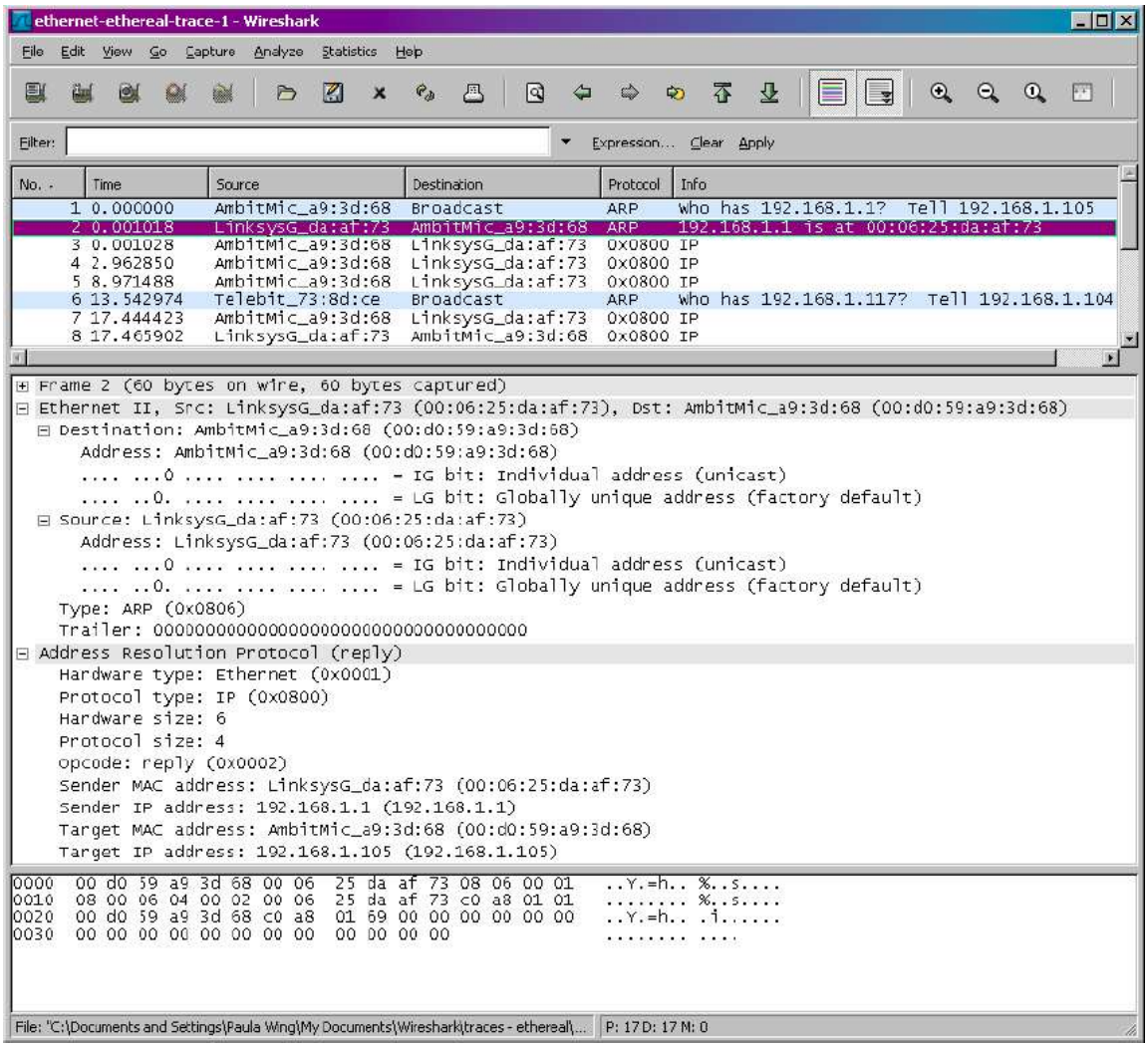
Above one shows the Ethernet address of my computer is 00:09:5b:61:8e:6d. The 48-bit destination address in the Ethernet frame is 00:0c:41:45:90:a8

Here is a screenshot showing the ARP request message



In the above one, the hex value for the source address is 00:d0:59:a9:3d:68. The hex value for the destination address is ff:ff:ff:ff:ff:ff, the broadcast address in the Ethernet frame. In the ARP packet Source MAC is 00:d0:59:a9:3d:68 and the Destination MAC is 0 since it is not known

Here is the screenshot for the ARP reply message



In the above one, the hex value for the source address is 00:06:25:da:af:73. The hex value for the destination address is 00:d0:59:a9:3d:68 in the Ethernet frame which means ARP reply is a unicast reply. ARP reply message also contains the same MAC address as in the Ethernet frame

Result:

Thus to capture and analyzes the packet traces of DHCP and ICMP protocol using Wireshark has been completed successfully.